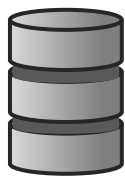
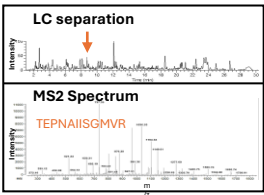


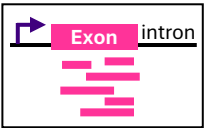
Public raw
LC-MS2
data



Proteome
database
search



GFF3-based
peptide
projection



Apollo
JBrowse
deployment



Genome-guided wheat proteogenomics workflow

1. DATA RETRIEVAL + REFERENCE GENOME + PROTEOME SEARCH

Steps 1–4

FTP download of raw LC-MS2 proteomic and GFF3 genomic files (Unix)

Data conversion (ProteoWizard MSConvert)

Protein database creation with decoy (Galaxy)

Proteome search against IWGSC v2.1 database (FragPipe: MSFragger → PeptideProphet → ProteinProphet)

2. GFF3 ANNOTATION FRAMEWORK

Step 5

ProteinID → TranscriptID → GeneModel → CDS blocks (Python)

3. PEPTIDE EVIDENCE CONSTRUCTION

Steps 6–8

FragPipe output importation (Python)

Non-contaminant peptide-protein-gene evidence (Python)

Protein to gene model projections using GFF3-derived table (Python)

4. GFF3-BASED PEPTIDE PROJECTION

Step 9

Peptide in protein → coding nucleotide coordinates → exon-aware genomic blocks (Python)

Protein sequence
MKWVTFPEPTIDEISLLFSSAY...
↓
Peptide CDS coordinates
nt 241–312
↓
GFF3 exon blocks
Exon1 – Intron – Exon2
↓
Genomic peptide projection
[redacted]

1. Locates the peptide amino acid sequence within the corresponding protein sequence.
2. Converts the peptide amino acid interval into coding nucleotide positions.
3. Retrieves the CDS blocks associated with the corresponding transcript.
4. Projects coding nucleotide positions onto genomic coordinates.
5. Collapses projected nucleotide positions into genomic blocks.
6. Generates an intron-aware peptide display label (exon-spanning junctions shown using dash characters).

5. PEPTIDE PROJECTION VALIDATION

Steps 10, 11

Translation validation: projected genomic blocks → translated peptide → match/mismatch QC (Python)

Independent sanity checks (Python)

Projected genomic blocks
[redacted]
↓
Retrieved genomic DNA
ATGGCC...GT...AAGTAA
↓
Translated peptide
NINGATTFSSALPVA
↓
Original peptide
= NINGATTFSSALPVA
↓
Evaluation
☑ match

Beside the translation validation, 3 independent sanity checks are performed:

1. BED geometry check: confirms BED coordinates by checking start coordinates < end coordinates, and that BED block sizes and block starts are consistent with each peptide genomic interval.
2. Chromosome and strand check: confirms that projected peptides are assigned only to expected wheat chromosomes and valid strand values (`+` or `-`).
3. Protein-coordinate consistency check: confirms that each peptide projection has coherent amino-acid coordinates, including matching peptide length and valid protein coordinate intervals.

Evaluation:

- translated sequence match + successful checks → peptide validated ☑
- translated sequence mismatch and/or unsuccessful check → peptide rejected ☒

6. RESOURCE OUTPUT + INTERPRETATION

Steps 12–25

validated projections exported as BED tracks (Python)

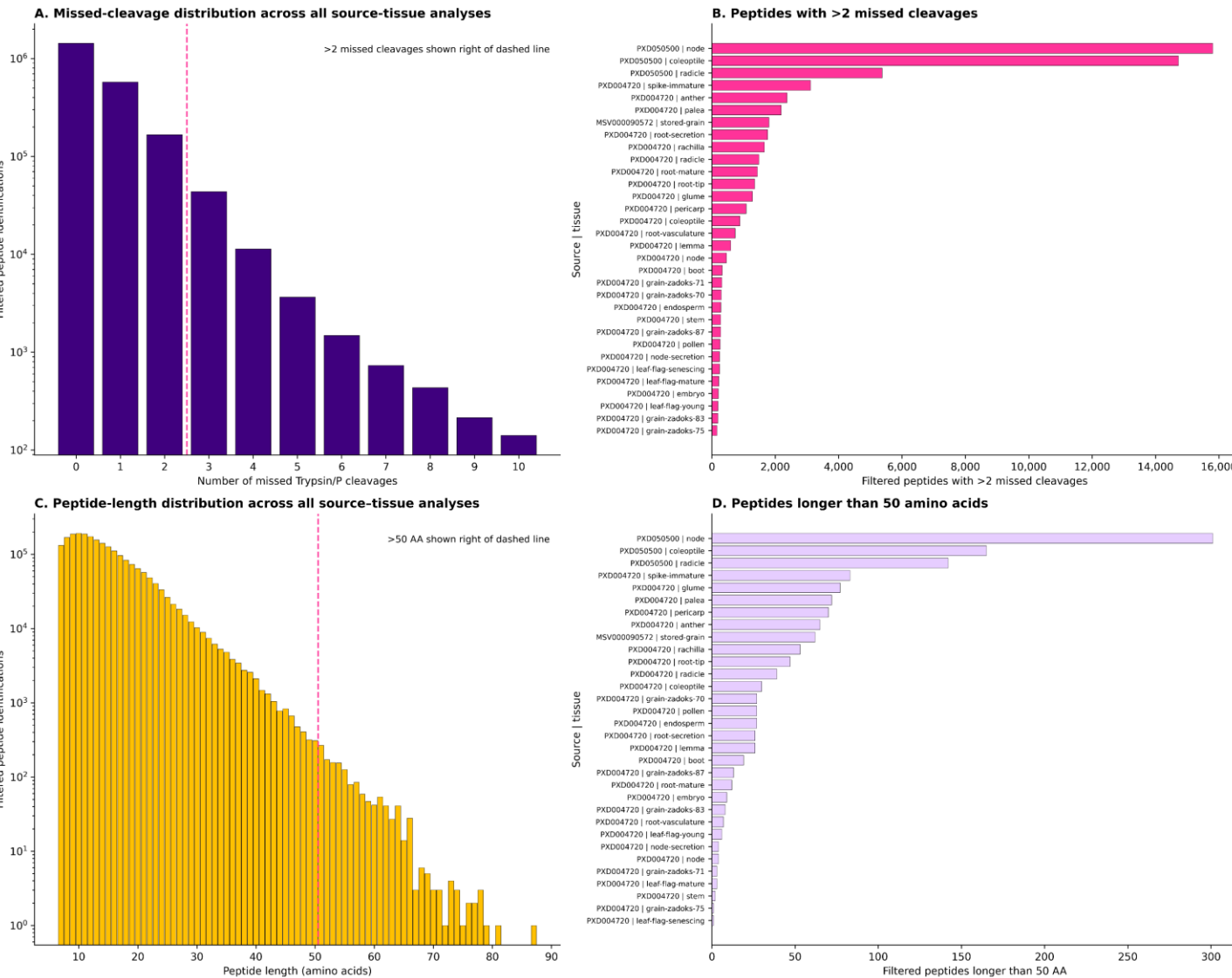
BED tracks permanent upload (Apollo Jbrowse)

EDA (scatter, box, violin, bar, UpSet, circular plots, histograms, Venn diagram, pie chart) (Python)

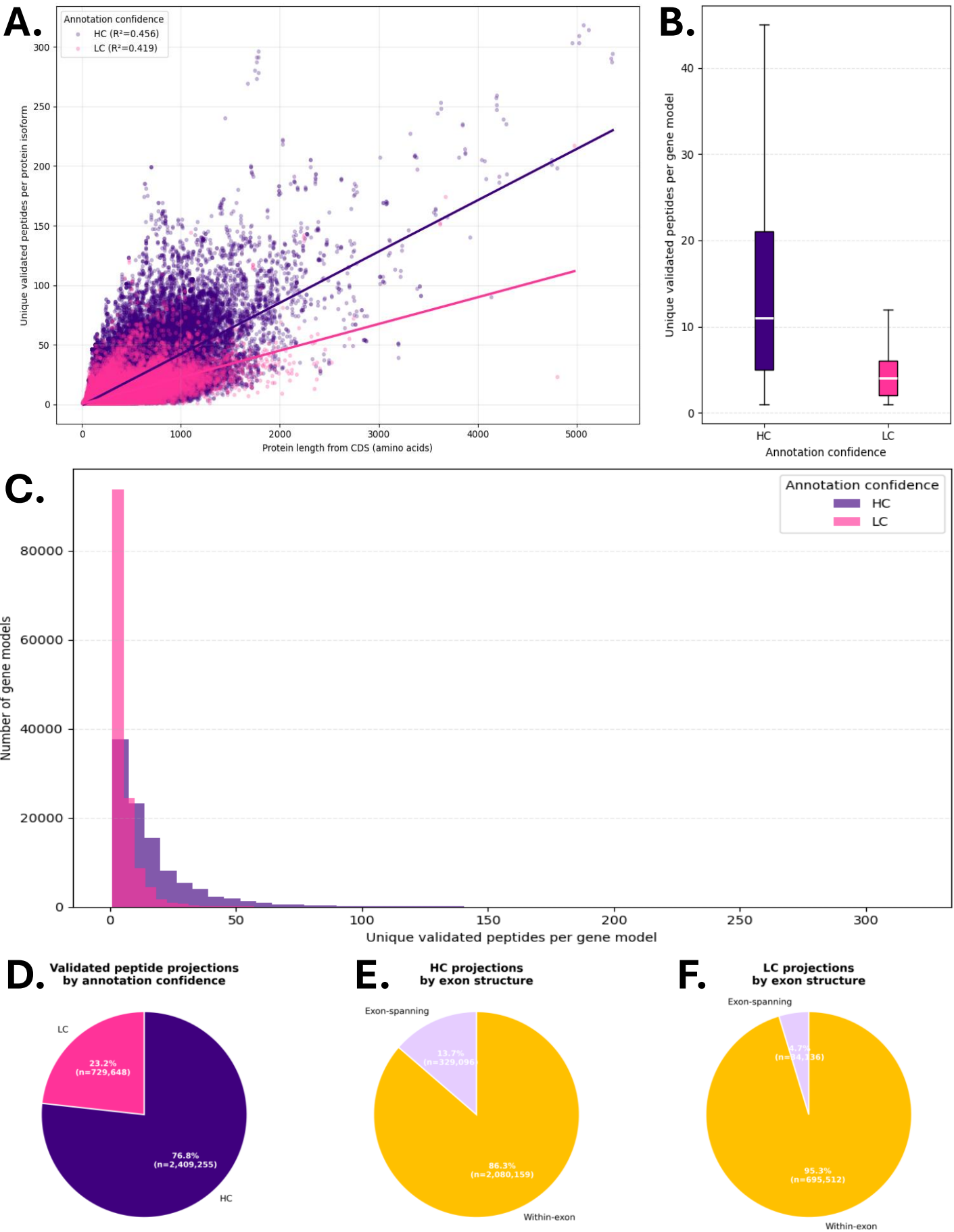
HC/LC support (Python)

Supplementary Figure S2: Empirical evidence supporting permissive missed-cleavage and peptide-length search limits used for FragPipe peptide identification. Distributions were calculated from the final FDR-filtered peptide.tsv reports generated independently for all 32 tissue-specific FragPipe analyses. Missed cleavages were determined from the peptide sequence assuming fully enzymatic Trypsin/P digestion, and peptide lengths were obtained directly from the FragPipe peptide reports. (A) Aggregate distribution of filtered peptide identifications according to the number of missed tryptic cleavages. The dashed line marks the conventional threshold of two miscleavages. Peptides containing up to ten miscleavages were retained, although their frequency decreased progressively with increasing missed-cleavage number. (B) Percentage of filtered peptide identifications with more than two missed cleavages for each tissue. High-miscleavage peptides were recovered across numerous tissues, demonstrating that the permissive digestion settings captured experimentally supported peptide classes rather than isolated outliers. (C) Aggregate distribution of peptide lengths across all tissues. The dashed line indicates peptides longer than 50 amino acids. Although uncommon, peptides substantially exceeding this length were identified, extending to approximately 90 amino acids in the combined dataset. (D) Percentage of filtered peptide identifications longer than 50 amino acids for each tissue. Long peptides were detected in multiple tissues, supporting the use of broad peptide-length limits during database searching. Together, these analyses demonstrate that the permissive search bounds recovered genuine peptide identifications that would likely have been excluded under narrower digestion and peptide-length constraints while remaining subject to subsequent PeptideProphet, ProteinProphet and Philosopher target-decoy validation.

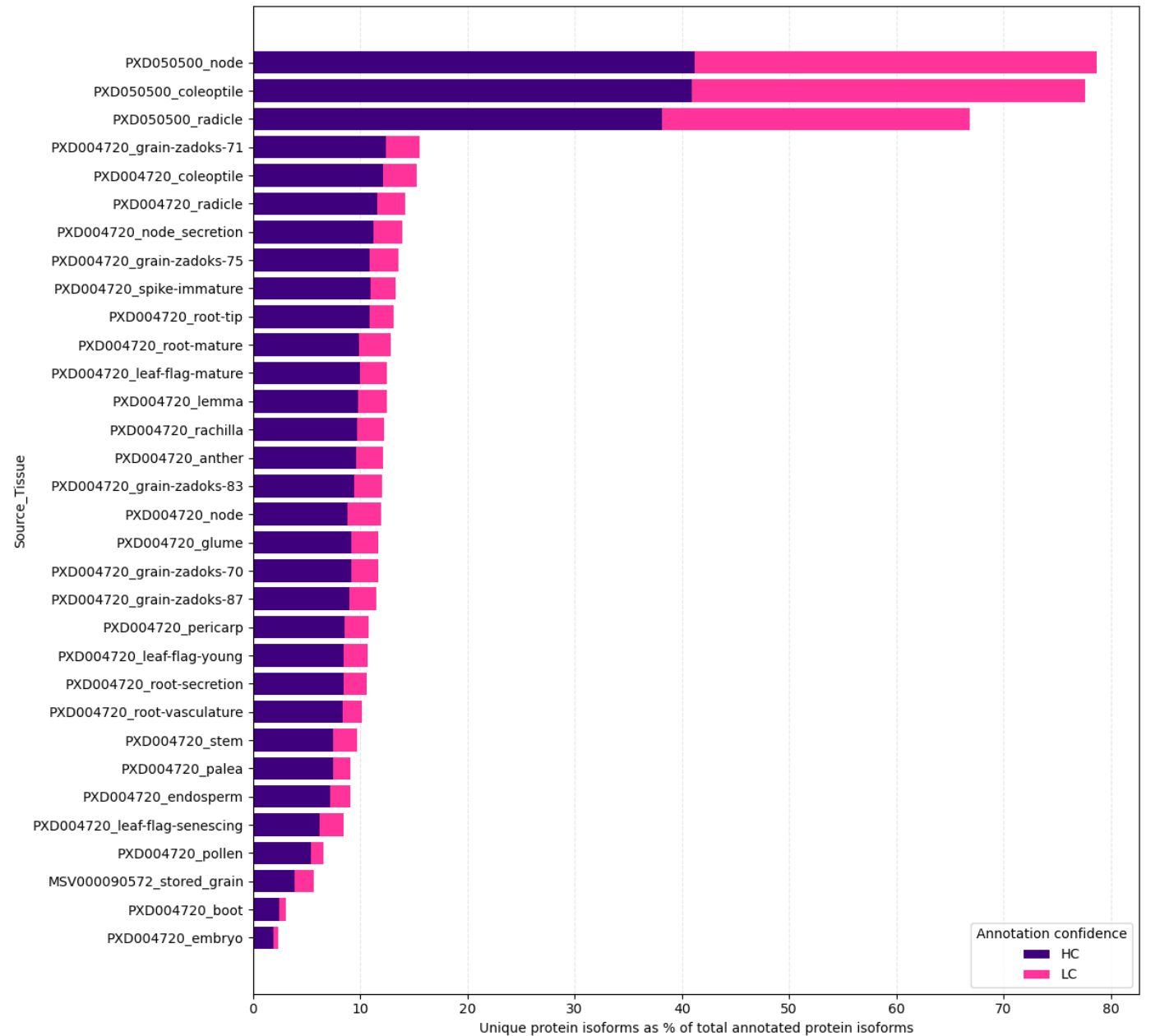
Empirical recovery of peptides enabled by permissive FragPipe digestion and peptide-length limits



Supplementary Figure S3: Peptide support distributions across HC and LC wheat gene models. (A) Scatterplot showing the relationship between protein length and the number of projected unique peptides per protein isoform, separated by HC and LC annotation confidence. (B) Boxplot comparing projected peptide support between HC and LC gene models. (C) Distribution of projected unique peptides per gene model showing broader and more abundant support for HC annotations. (D) Proportion of non-redundant validated peptide projections assigned to HC and LC gene models. (E) Proportion of HC peptide projections classified as within-exon or exon-spanning. (F) Proportion of LC peptide projections classified as within-exon or exon-spanning.



Supplementary Figure S4. Proteogenomic coverage across wheat tissues. Barplot of protein-level coverage showing the percentage of annotated HC and LC gene models supported by projected protein accessions.



Supplementary Figure S5: UpSet plot of projected protein intersections across wheat tissues. Vertical bars indicate the number of shared projected proteins among tissue combinations, while horizontal bars show the total number of projected proteins identified per tissue. The analysis highlights both broadly shared and tissue-specific proteogenomic signatures across the wheat datasets.

